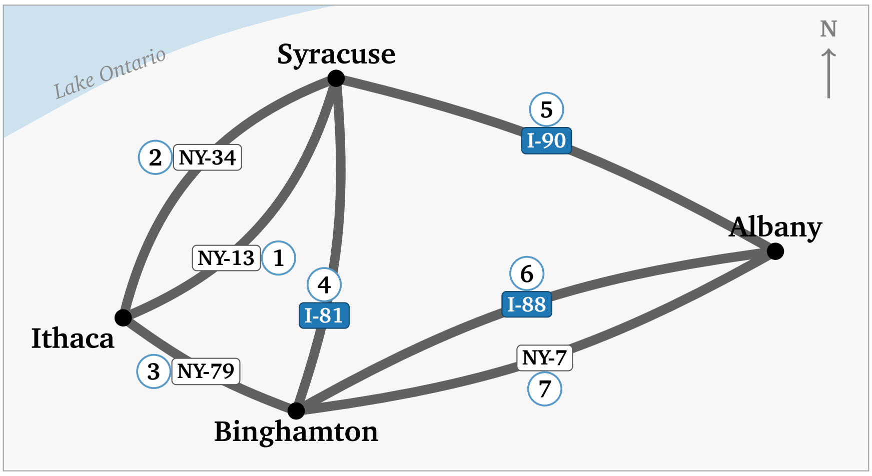
Answers

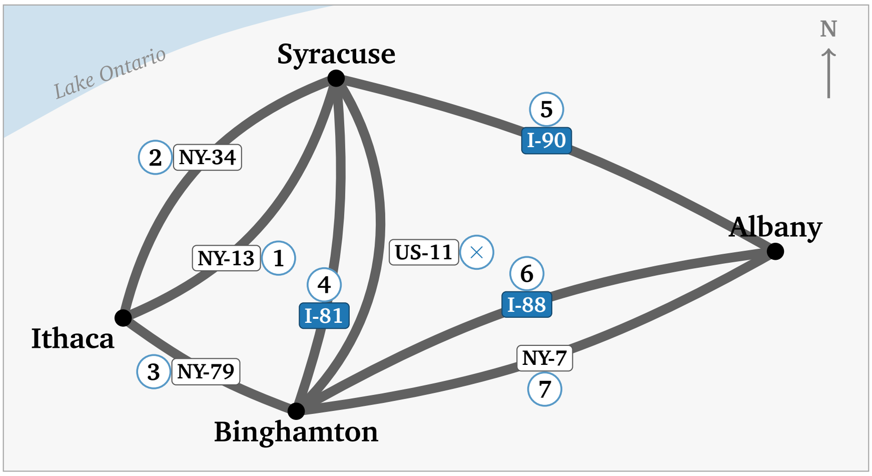
Every question, with the reasoning worth saying out loud. Question 1(a) has many right answers; the rest have one.

Part 1 — Seven highways

1(a): One drive out of many. It has to start at **Ithaca** and finish at **Albany**, whichever roads it takes in between — those are the two cities with an odd number of roads, and an odd city is where a drive begins or ends.



1(b): **No.** Seven of the eight can be driven — the numbers above still work — and **US-11 is left over**, or I-81 if you would rather leave that one. US-11 pushes Syracuse to five roads and Binghamton to five, so **four** cities now have a left-over road, and a drive has one start and one finish.



2:

	2 roads	3 roads	4 roads	5 roads
pairs you drew	1	1	2	2
roads left over	0	1	0	1

(a) **2 and 4** — the ones with nothing left over. (b) **They are even.** (c) **start or finish.**

3:

	I	S	B	A	how many have a left-over?
seven roads	3	4	4	3	2
with US-11 too	3	5	5	3	4

At most **2** cities may have a left-over road. Seven roads: **a drive exists** (Ithaca to Albany). Eight roads: **none exists**. No route with eight because **four cities have a left-over road, and a drive has only one start and one finish.**

4: Königsberg. Bridges: **A 5, B 3, C 3, D 3** — fourteen ends for seven bridges, as always. The walk is **not possible**: all four pieces of land are odd.

Destroying one bridge: **any of the seven works.** Every bridge joins two odd lands, so knocking one out makes both of them even and leaves exactly two odd lands — and the map stays in one piece whichever you choose. (Königsberg's own answer was to build one, not to destroy one.)

Part 2 — Three kinds of route

5(a):

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	**walk**
A B D C B	no	yes (B)	**trail**
A B C D E	no	no	**path**

5(b): **A D B C D E.** The lines A–D, D–B, B–C, C–D, D–E are five different lines; D is the stop that comes round twice.

5(c): **a trail.** No road twice makes it a trail; coming back to a city it had already visited is what stops it being a path.

5(d): A path of 7 roads would touch **8** different cities — one more city than roads, because it never comes back — and the highway map has only **4**.

Part 3 — Predictions for the notebook

6(a) and **6(b)**:

$$A = \begin{array}{c|cccc|c} & 1 & 2 & 3 & 4 & \text{row total} \\ \hline 1 & 0 & 1 & 0 & 0 & \mathbf{1} \\ 2 & 1 & 0 & 1 & 1 & \mathbf{3} \\ 3 & 0 & 1 & 0 & 1 & \mathbf{2} \\ 4 & 0 & 1 & 1 & 0 & \mathbf{2} \end{array}$$

Each row total is the **degree** of that node.

6(c): from 1 to 3: **1** of them, 1 → **2** → 3. From 1 to 2: **0** — **the first step has to go to 2, and the second step has to leave it again.** From node 2 back to node 2: **3** — out and back along each of its three lines.

6(d): row 1, column 3: **1.** row 1, column 2: **0.** The diagonal is **the degrees** — 1, 3, 2, 2 — because a 2-step route that comes back is one line out and the same line home.

$$A^2 = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Part 4 — The lab

The seven roads, read off the map — order does not matter, but both Ithaca–Syracuse roads must be there and both Binghamton–Albany roads must be there:

```
ROADS = [(0, 1), (0, 1), (0, 2), (1, 2), (1, 3), (2, 3), (2, 3)]

def to_adjacency(edges, n):
    A = np.zeros((n, n), dtype=int)
    for i, j in edges:
        A[i, j] += 1
        A[j, i] += 1
    return A

def degrees(A):
    return A.sum(axis=1)

def euler_status(A):
    if not is_connected(A):
        return "impossible"
    odd = int(np.sum(degrees(A) % 2 == 1))
    if odd == 0:
        return "circuit"
    if odd == 2:
        return "path"
    return "impossible"
```

The verdicts: seven roads **path**, with US-11 **impossible** — 1(a) and 1(b) again, from a rule that never looked at the picture.

Finished early. Every degree even and still no tour means the map is **in more than one piece**: two triangles, say,

```
CHALLENGE = [(0, 1), (1, 2), (2, 0), (3, 4), (4, 5), (5, 3)]
```

Every place has two roads, and no drive gets from one triangle to the other. A rule that only counted parity says `circuit` here and is wrong; that is what `is_connected(A)` is for. One road joining the two triangles makes its two ends odd and the answer `path`; a second road *beside the first*, between the same two places, makes everything even again and the answer `circuit`.